

General Knowledge

Indian Geography

STUDY NOTES

**Mineral Resources in India - Coal, Petroleum
and Natural Gas**



MINERAL RESOURCES IN INDIA: COAL, PETROLEUM & NATURAL GAS

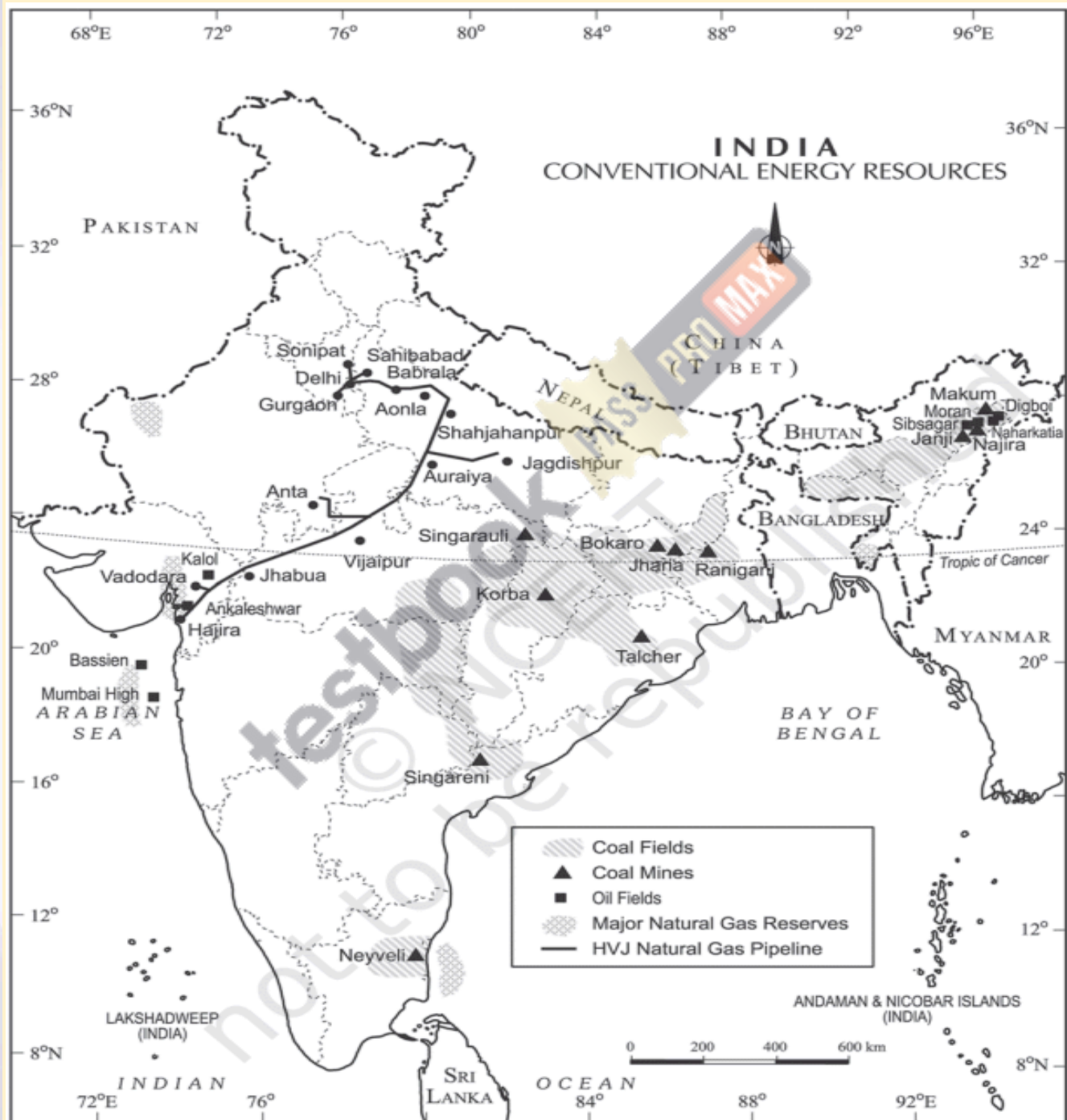


Fig. 7.5 : India – Conventional Energy Resources

Coal: A Vital Energy Resource in India

1. Introduction

- **Coal** is often called “**black gold**” and forms the backbone of India’s energy infrastructure, as it is the most abundant fossil fuel in the country.
- It is primarily used for **power generation** and is crucial for **industrial sectors** like **iron and steel** manufacturing. In India, coal provides nearly **70%** of the total electricity production.

2. Type of Resource

- **Fossil Fuel / Conventional Energy Source**
- Originates from **decomposed plant material** under layers of soil over millions of years.
- Major types are based on **carbon content** and include **peat**, **lignite**, **bituminous**, and **anthracite**.

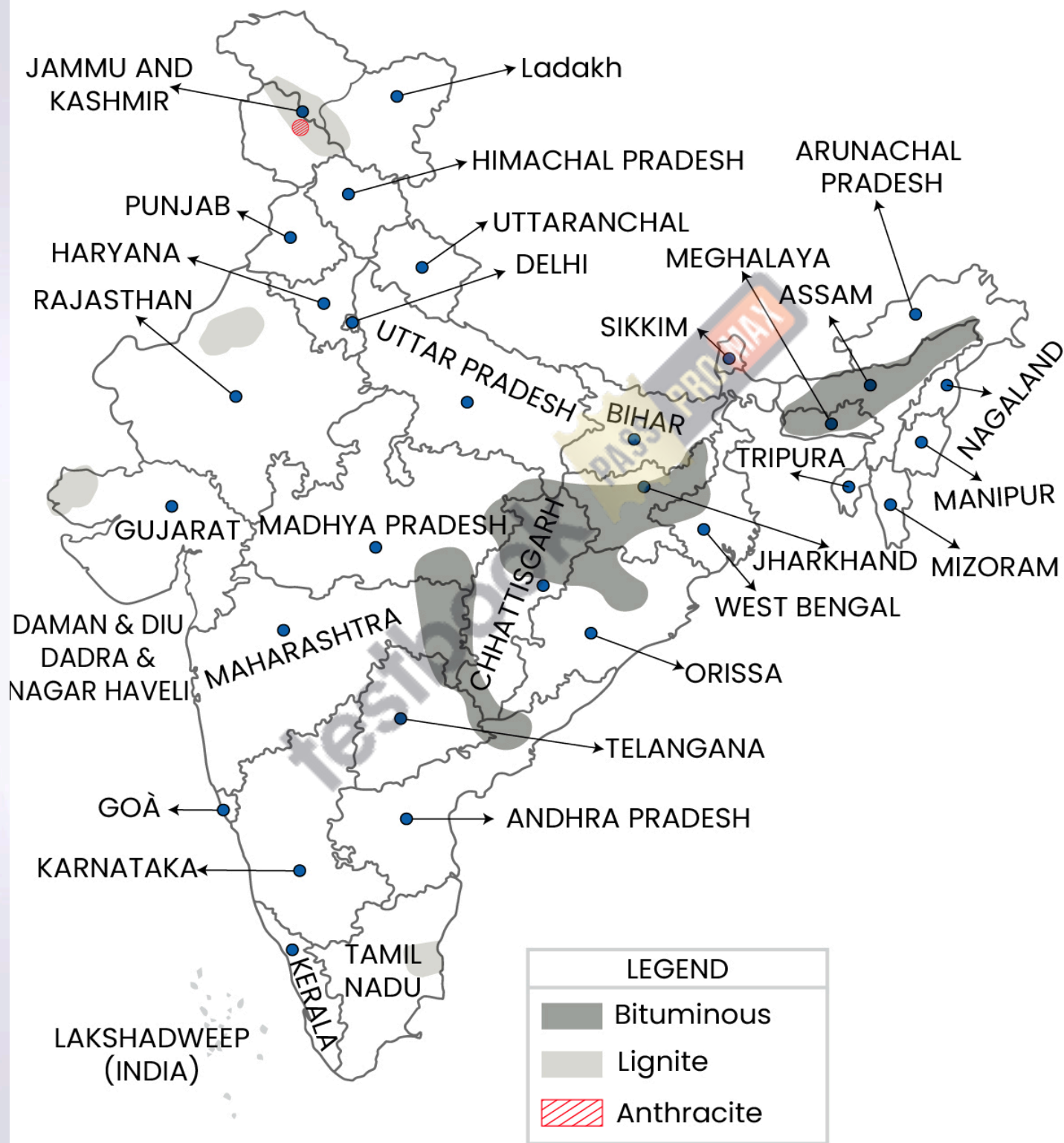
3. Formation Process

- Formed through the **carbonization** of **plant material** over **millions of years**.
- Over time, the proportion of **carbon** increases as **oxygen and nitrogen decrease** due to **heat and pressure**.
- **Coalification stages**:
 - **Peat**: Initial accumulation of plant debris.
 - **Lignite**: Low-grade Brown coal with moderate carbon content.
 - **Bituminous**: Soft coal, widely used with 40-80% carbon.
 - **Anthracite**: Hard coal with the highest carbon content (80-95%), provides high energy.

4. Types and Classifications

- **Peat**: Contains <40% carbon, has low energy output and produces more smoke.
- **Lignite (Brown Coal)**: 40-55% carbon, high moisture, used in electricity generation (notably in **Neyveli, Tamil Nadu**).
- **Bituminous Coal**: Contains 45-86% carbon, high calorific value, most widely used for **coke and gas production**.
- **Anthracite**: 86-97% carbon, found in **Jammu & Kashmir**, high calorific value and efficiency.

Coal Reserves in India



5. Distribution in India

- **Gondwana Coal Fields** (about 200 million years old) and **Tertiary Coal Fields** (around 55 million years old).
- **Major Coal Reserves:**
 - **Damodar Valley** (Jharkhand–West Bengal): Contains the **Raniganj, Jharia, Bokaro, Giridih, and Karanpura** coalfields.
 - **Mahanadi Valley** (Chhattisgarh): Known for **Korba coalfield**.
 - **Godavari Valley** (Telangana): Home to **Singareni** and other deposits.
- **Lignite Deposits:** Found in **Tamil Nadu, Puducherry, Gujarat, and Jammu & Kashmir**.
- **Key Coal Mining Centers:**
 - **Jharkhand:** Jharia, Bokaro
 - **Chhattisgarh:** Korba, Rampur
 - **Odisha:** Talcher
 - **Maharashtra:** Chanda–Wardha
 - **Telangana:** Singareni
- **Tertiary Coal Fields:**
 - **Northeastern States:** Assam, Meghalaya, Nagaland, and Arunachal Pradesh.

6. Uses and Applications

- **Power Generation:** Nearly **70%** of India's electricity comes from coal-fired thermal power plants.
- **Iron and Steel Industry:** Used as **coking coal** in iron smelting processes.
- **Domestic and Industrial Use:** In households and smaller industries for **heating and fuel**.
- **By-products:** Coal tar, coal gas, and coke, used in various **chemical and industrial applications**.

7. Environmental Impact and Challenges

- **High Ash Content:** Indian coal has high ash content, which leads to air pollution and health issues.

- **Air and Water Pollution:** Coal combustion releases **CO₂, SO₂, and NO_x**, contributing to **climate change** and **acid rain**.
- **Deforestation and Land Degradation:** Mining disrupts ecosystems and leads to **soil erosion**.
- **Water Stress:** Coal mining consumes significant water resources, impacting availability in surrounding areas.
- **Rat Hole Mining:** A hazardous and environmentally damaging mining method in Meghalaya.

8. Economic Importance

- **Employment:** Major employer in mining regions, providing jobs for thousands.
- **Revenue Generation:** Contributes significantly to **India's GDP** through taxes and royalties.
- **Energy Security:** Coal is India's primary energy source, reducing dependency on imported energy.

9. Recent Developments and Government Initiatives

- **Commercial Coal Mining:** In 2020, India **allowed** private firms to enter the coal sector, aiming to increase production and reduce imports.
- **Coal Block Auctions:** Auctions held to allocate mining blocks for private and public sectors to increase efficiency.
- **Clean Coal Technologies:** Emphasis on **coal gasification** and **carbon capture and storage (CCS)** to reduce environmental impact.
- **Increased Lignite Use:** Tamil Nadu's Neyveli Lignite Corporation focuses on expanding lignite mining for regional power needs.

10. Conclusion

- Coal remains a **critical energy resource** for India's growth, especially in **power generation** and **industry**.
- Moving forward, **environmental sustainability** and **clean coal technologies** will be essential to mitigate coal's environmental impact while meeting energy needs.

Petroleum: Liquid Gold and Key Energy Source in India

1. Introduction

- **Petroleum**, often referred to as “**liquid gold**”, is a crucial fossil fuel primarily used for **transportation, power generation, and industrial applications**.
- Formed from **marine organisms** buried under sedimentary rock layers over millions of years, petroleum is indispensable to India’s energy security.

2. Type of Resource

- **Fossil Fuel / Conventional Energy Source**
- Found in **sedimentary basins** and **offshore reserves**.
- Includes **crude oil, natural gas**, and by-products processed in **petrochemical industries**.

3. Formation Process

- Derived from **organic materials** like plankton and algae that settled in ancient seas.
- **Heat and pressure** over millions of years convert this material into hydrocarbons.
- **Oil traps** in sedimentary rock layers hold petroleum, which accumulates in **anticlines, fault traps, and salt domes**.

4. Types and Classifications

- **Crude Oil**: Unrefined oil containing a mixture of hydrocarbons.
- **Associated Gas**: Natural gas found with crude oil in reservoirs.
- **Non-Associated Gas**: Natural gas found in separate reservoirs from oil.
- **Sweet Crude**: Low in sulfur, easier to refine.
- **Sour Crude**: High in sulfur, requires additional refining.

5. Distribution in India

- **Key Petroleum Regions:**
 - **Mumbai High** (Maharashtra): Offshore field, 160 km off the coast, discovered in 1973, accounting for major production.
 - **Gujarat**: Major oilfields at **Ankleshwar, Kalol, Mehsana, Nawagam, and Kosamba**.
 - **Assam**: The oldest oil-producing region, with key fields at **Digboi, Naharkatiya, and Moran**.

- **Krishna-Godavari Basin** (Andhra Pradesh and Tamil Nadu): Important discoveries in offshore reserves on the east coast.
- **Other Significant Reserves:**
 - **Cauvery Basin** (Tamil Nadu)
 - **Rajasthan:** Notable onshore reserves in the **Barmer Basin**.
- **Strategic Petroleum Reserve:**
 - Facilities in **Visakhapatnam (Andhra Pradesh)**, **Mangalore**, and **Padur (Karnataka)** provide emergency reserves.

6. Petroleum Refining in India

- **First Refinery:** Established in **1901** at **Digboi, Assam**, marking the beginning of India's petroleum refining.
- **Expansion:** An additional refinery started in **1954** at **Tarapur, Mumbai**.
- **Current Refining Hub:** India now operates as a **refining hub**, with refining capacity exceeding domestic demand, enabling **exports of refined petroleum products**.
- **Transportation of Crude Oil: Pipelines** are used to transport crude oil from wells to the nearest refineries.

7. Advantages of Pipelines

- **Efficient for Liquids and Gases:** Ideal for transporting petroleum, natural gas, and other liquids.
- **Challenging Terrains:** Can be laid across difficult terrains and **underwater**, enhancing connectivity.
- **Low Maintenance:** Requires minimal maintenance compared to other transportation methods.
- **Safe and Environmentally Friendly:** Offers accident-free, environmentally sustainable transportation.

8. Disadvantages of Pipelines

- **Limited Flexibility:** Pipelines connect only fixed points, reducing adaptability.
- **Capacity Constraints:** Once laid, the capacity of a pipeline cannot easily be increased.
- **Security Concerns:** Hard to ensure security over long distances.

- **Leak Detection and Repair:** Difficulties in leak detection and repairs due to pipeline structure.

9. Major Crude Oil Pipelines in India

- **Salaya-Mathura Pipeline (SMPL)**
- **Paradip-Haldia-Barauni Pipeline (PHBPL)**
- **Mundra-Panipat Pipeline (MPPL)**

10. Key Petroleum Product Pipelines

- **Guwahati-Siliguri Pipeline (GSPL)**
- **Koyali-Ahmedabad Pipeline (KAPL)**
- **Barauni-Kanpur Pipeline (BKPL)**
- **Panipat-Delhi Pipeline (PDPL)**
- **Panipat-Rewari Pipeline (PRPL)**
- **Chennai-Trichy-Madurai Product Pipeline (CTMPL)**
- **Chennai-Bangalore Pipeline**

11. Additional Significant Pipelines

- **Naharkatia-Nunmati-Barauni Pipeline:** India's first pipeline.
- **Mumbai High-Mumbai-Ankleshwar-Koyali Pipeline**
- **Hajira-Bijapur-Jagdishpur (HBJ) Gas Pipeline**
- **Jamnagar-Loni LPG Pipeline**
- **Kochi-Mangalore-Bangalore Pipeline**
- **Visakhapatnam-Secunderabad Pipeline**
- **Mangalore-Chennai Pipeline**
- **Vijayawada-Visakhapatnam Pipeline**

12. Uses and Applications

- **Transportation Fuel:** Fuels cars, trucks, aircraft, and ships with gasoline, diesel, and aviation fuel.
- **Power Generation:** Used in diesel-based power plants and as an emergency fuel source.

- **Petrochemical Industry:** Source for **fertilizers, synthetic rubber, synthetic fibers, plastics, and medicines.**
- **Industrial Applications:** Used in **lubricants, waxes, and cosmetics.**

13. Environmental Impact and Challenges

- **Greenhouse Gas Emissions:** Petroleum combustion releases **CO₂**, a key contributor to **global warming.**
- **Oil Spills and Water Pollution:** Leaks in drilling and transportation cause significant **marine pollution.**
- **Deforestation and Habitat Loss:** Exploration activities impact local ecosystems.
- **Air Pollution:** Emissions include **SO₂, NO_x**, and particulate matter, affecting air quality and human health.

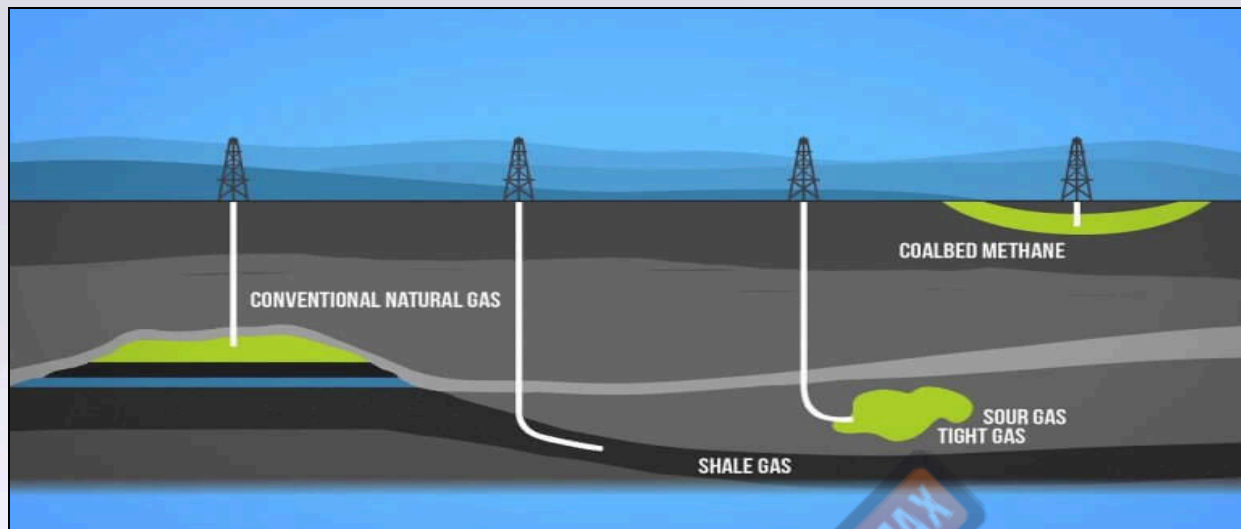
14. Economic Importance

- **Energy Security:** Ensures India's fuel independence and is a key component of **strategic reserves.**
- **Revenue Generation:** Contributes significantly to the economy through **excise duty, royalties, and exports.**
- **Employment:** Provides jobs in **drilling, refining, transportation, and petrochemical industries.**
- **Foreign Exchange:** Imports of petroleum products have a major impact on India's balance of payments.

15. Recent Developments and Government Initiatives

- **New Discoveries:** Exploration in **Krishna-Godavari Basin, Rajasthan's Barmer Basin, and Cauvery Basin.**
- **Strategic Petroleum Reserves (SPR):** Expansion planned for **Chandikhole (Odisha) and Bikaner (Rajasthan).**
- **Enhanced Oil Recovery (EOR) Techniques:** Government emphasis on advanced methods like **carbon dioxide injection** to boost output.
- **Energy Transition Initiatives:** Diversifying from oil to **renewable sources and natural gas** to meet environmental targets.

Natural Gas: India's Strategic and Environmental Fuel



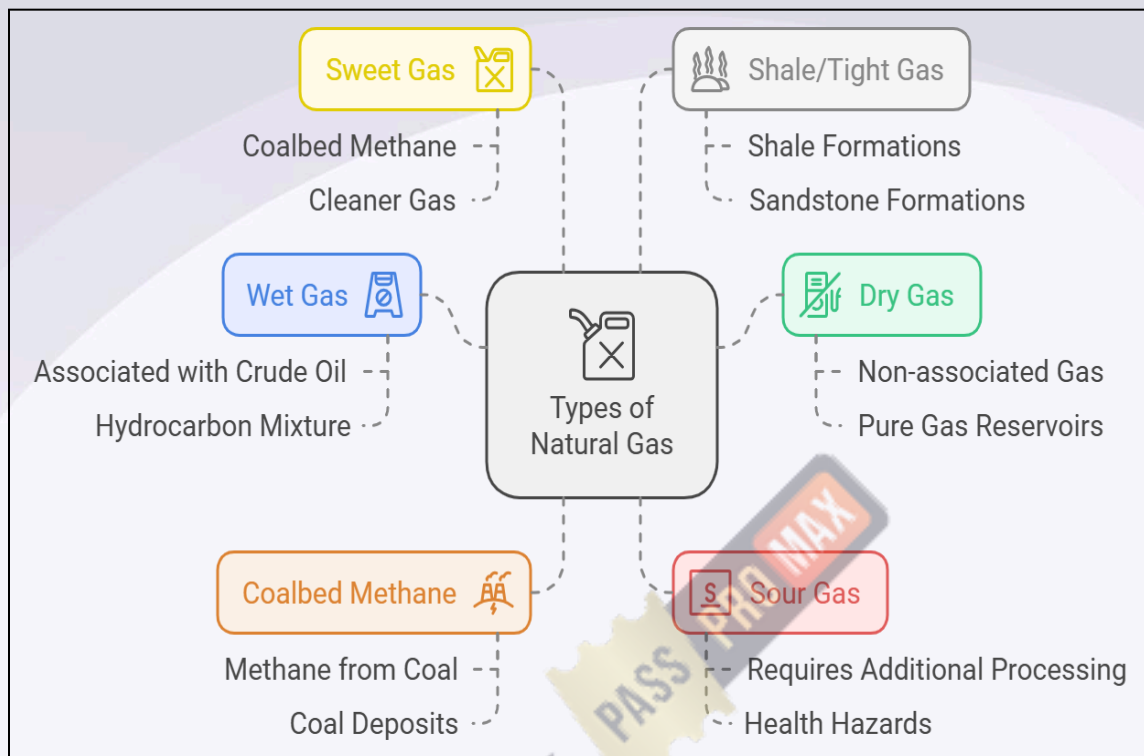
1. Introduction

- **Natural gas** is a **fossil fuel** primarily composed of **methane**, formed from the decomposed organic matter of plants and animals under immense heat and pressure over millions of years. It is found deep underground, often near **coal and crude oil beds**.
- **Importance:** Natural gas is vital for **power generation, domestic use, and industrial processes**, offering a cleaner alternative to coal and oil. It is a **fossil fuel**, primarily composed of **methane**, extracted from deep earth reservoirs near coal and crude oil beds.
- **Gas Authority of India Limited (GAIL)**
 - **Foundation:** Established in **1984**.
 - **Role:** India's **largest natural gas company** and one of the **seven Maharatna Public Sector Undertakings (PSUs)**.
 - **Ministry:** Operates under the **Ministry of Petroleum and Natural Gas**.

2. Type of Energy Source

- **Classification:** Natural gas is a **non-renewable fossil fuel**.
- **Formation:** Created over millions of years from **decomposed organic matter** subjected to **high pressure and heat** deep within the earth.
- **By-products:** During processing, it yields **propane, ethane, butane, carbon dioxide, and nitrogen**.

3. Types of Natural Gas



4. Major Producing Regions in India

- **West Coast: Mumbai High** and surrounding areas in Gujarat and Maharashtra.
- **East Coast:** New reserves in **Krishna-Godavari Basin** and **Cauvery Basin**.
- **Other Notable Regions:** Assam Shelf, Tripura, and Barmer Basin in Rajasthan.

5. Uses of Natural Gas

- **Power Generation:** Fuels about 15% of India's electricity needs (as of 2023).
- **Industrial Use:** Powers boilers, and heating systems, and serves as raw material in **fertilizer and chemical industries**.
- **Transportation:** Used as **Compressed Natural Gas (CNG)** for vehicles.
- **Domestic Use:** **Piped Natural Gas (PNG)** for cooking and heating in homes.

6. Benefits of Natural Gas as Fuel

- **Cleaner Fuel:** Emits fewer pollutants and greenhouse gases compared to coal and oil.

- **Efficient:** High energy conversion efficiency for maximum utilization.
- **Versatile:** Used across multiple sectors like **cooking, electricity generation, and transportation.**
- **Flexible:** Can be transported and stored in liquefied (LNG) or compressed (CNG) forms.

8. Storage of Natural Gas

- **Underground Storage:** Utilizes **depleted gas fields** and **salt caverns** for large-scale storage.
- **Above-Ground Storage:** Designed for smaller-scale reserves.
- **Advantages of Underground Storage:**
 - **Capacity:** Supports extensive storage.
 - **Energy Efficiency:** Maintains gas in a **condensed state**, minimizing energy losses.
 - **Cost-Effective:** Low maintenance costs.
 - **Safety:** Reduces leakage risk and ensures environmental protection.
 - **Grid Reliability:** Helps balance supply during **peak demand**.

9. Forms of Natural Gas for Distribution and Use

- **Liquefied Natural Gas (LNG):** Cooled to -160°C for large-volume transport in cryogenic tanks.
- **Regasified Liquefied Natural Gas (RLNG):** LNG is converted back to gas form before distribution.
- **Compressed Natural Gas (CNG):** Natural gas compressed to $200\text{-}250\text{ kg/cm}^2$, used mainly in **vehicles**.
- **Piped Natural Gas (PNG):** Distributed through pipelines, ideal for **domestic cooking and heating**.